

High Data Rate Receiver with Inbuilt Modulator

High data Rate Receiver modem is a high performing system for receiving & processing satellite or aircraft data. It is deployed most commonly within a satellite ground station with data reception and processing at high-rate signals from LEO or GEO Satellites. It is configurable to accommodate multiple demodulation schemes & decoding algorithms. It is suitable for high data rate scientific, remote sensing and telecommunications applications.

Introduction

The High Data Rate Receiver (HDR) with inbuilt modulator is an advanced communication system designed for receiving, processing, and transmitting high-speed satellite or airborne data signals. It is primarily deployed in satellite ground stations to support data reception from LEO and GEO satellites and offers flexible configuration to support multiple demodulation schemes and decoding algorithms for diverse mission requirements.

Built on a fully digital, FPGA-based architecture, the system enables high symbol rate operation, dual-channel independent processing, and efficient baseband signal handling. It integrates advanced forward error correction, synchronization, and real-time processing capabilities to ensure reliable data recovery even under challenging signal conditions.

The HDR also includes an inbuilt modulator with channel emulation capability, high-speed Ethernet and LVDS outputs, built-in storage, and real-time monitoring features. These capabilities make it well suited for high-data-rate applications such as remote sensing, scientific missions, telemetry processing, and telecommunications.



Architecture

The High Data Rate Receiver (HDR) is based on a fully digital, FPGA-centric architecture designed to handle high-speed signal reception, processing, and transmission efficiently. The system incorporates dual independent IF channels that can operate simultaneously without performance degradation, enabling parallel reception and processing of multiple high-rate signals. All major baseband functions, including demodulation, synchronization, decoding, and post-processing, are implemented within a high-end FPGA to ensure real-time operation and flexibility.

The architecture integrates configurable demodulators, advanced forward error correction modules, frame synchronization, and data formatting blocks, along with high-speed Ethernet and LVDS interfaces for output. In addition, it includes an inbuilt modulator and channel emulation capability, allowing both signal reception and transmission within a single platform, making it a comprehensive end-to-end high-data-rate communication solution.

Why Choose Us?

-  Rapid integration, faster time-to-market
-  Clean, verified, documented RTL
-  Direct access to experts



Features

- Fully digital, tunable transmitter and receiver
- 720MHz with 320MHz Symbol Rate, 2.4GHz IF with 1000 MHz Symbol Rate
- 2 IF-Channels (independent)
- 8 channels of LVDS line/ECL for I/O
- Receives signals of power level -10 dBm to -50 dBm
- BPSK, QPSK, UQPSK, OQPSK, 8PSK selectable demodulators
- Viterbi decoding, differential decoding, 4D-TCM, Reed Solomon decoder, LDPC Decoding, SCCC Decoding
- Optional Frame Synchronization, CADU, VCDU Decoding & Bit error measurement
- Ethernet and LVDS data output provided with inbuilt Storage
- Inbuilt Modulator with Channel Emulation Capability
- High end FPGA with entire base-band processing ported in it.
- Supports multiple data formats for display.
- Built-in support for real time debugging.
- Compliant with CCSDS recommendations

Dual-Channel IF Receiver and Demodulator Specification

IF Input	
Parameter	Specification
No of IF Inputs	• 2
Input Frequency	• 720MHz, 2.4GHz
Max Symbol Rate @ 720MHz	• 320MSps
Max Symbol Rate @ 2.4GHz	• 1000MSps
IF Tunability @ 720MHz	• 500MHz to 940MHz
IF Tunability @ 2.4GHz	• 1.6GHz to 3.15GHz
IF Tunability	• ± 100 KHz step size
Power	• -10dBm to -50 dBm
VSWR	• ≤ 1.5
IF Connector Type	• SMA Female



Acquisition Range	Up-to ± 1 MHz / Auto
Doppler Rate	20KHz/sec
Acquisition Time	<200ms
Notes: - Both IF channels can operate independently and parallelly without any degradation in BER performance. - The Max Symbol Rate can be upgraded to up-to 500MSps symbol rate @ 720MHz.	
DEMODULATION & BIT-SYNC	
Parameter	Specification
Supported schemes	BPSK, QPSK, UQPSK, OQPSK, 8PSK (Selectable)
Spectral Inversion support	I&Q Swap, $\bar{I}\bar{Q}$, $\bar{I}Q$, $I\bar{Q}$
Symbol Rate	10 - 1000 MSPS
Max Bit rate @720MHz	- BPSK (1 X 320) Mbps - QPSK/UQPSK/OQPSK- (2X320) Mbps 8PSK (3 X 320) Mbps
Max Bit rate @2.4GHz	- BPSK (1 X 1000) Mbps - QPSK/UQPSK/OQPSK-(2X1000) Mbps, 8PSK (3 X 1000) Mbps
BER Degradation	- BPSK / QPSK: 1 dB @ 10^{-6} BER - UQPSK / OQPSK / 8PSK: < 1.5 dB @ 10^{-6} BER
Equalization Filtering	Digital Adaptive Equalization (Automatic or Manual)
Cross Polarization Cancellation	Selectable (Enable/Disable)
Phase Ambiguity Resolution	Automatic or Manual
DECODING	
Parameter	Specification
Viterbi Decoding	- Selectable rate: 1/2 punctured to 2/3, 3/4, 5/6, and 7/8 - Max symbol rate in merged mode: 320 MSps - Max symbol rate in multi-channel mode: 640 MSps - Option of Enable/Disable Viterbi Decoding
Differential Decoder	- Max symbol rate – 1000MSps - Option of Enable/Disable Differential Decoding



Reed Solomon Decoder	- Compliant to CCSDS recommendation (as per CCSDS 131.0- B-2 Aug 2011). Following RS Decoders are supported. (i) CCSDS RS (223,255) (ii) CCSDS RS (239,255) - Additionally, CCSDS RS (247,255) is also available on request. - Option of shortening (Virtual fill up-to 16 bytes) and Interleaving depth of 1 to 8 Code-words. - Max Data-Rate 2000MSps - Option of Enable/Disable RS Decoding
LDPC Decoder	- Compliant to CCSDS 131.1-O-2 - Code-Rate of $\frac{1}{2}$, $\frac{2}{3}$, $\frac{4}{5}$, $\frac{7}{8}$ (Max Symbol Rate 500MSps) - Note - $\frac{7}{8}$ Code-rate can operate at maximum 2000Mbps full speed. (i) $\frac{7}{8}$ – Shortened (8160, 7136) code (ii) $\frac{1}{2}$ - 1K, 4K block length (iii) $\frac{2}{3}$ – 4K block length (iv) $\frac{4}{5}$ – 4K block length - Option of Enable/Disable LDPC Decoding - Note - code rate of $\frac{1}{2}$, $\frac{2}{3}$, $\frac{4}{5}$ available as SW upgradation option.
8PSK 4D TCM Decoder	- Output rate 2, 2.25, 2.5 and 2.75 - Option of Enable/Disable TCM Decoding
DVB-S2 Support	Optional, as per CCSDS specs 131.3-B2, Adaptive Coding and Modulation (ACM) mode is also supported.
DVB-S2x Support	Optional
CCSDS SCCC Support	Optional, Adaptive Coding and Modulation (ACM) mode is also supported.
FRAME SYNC & DE-RANDOMIZATION	
Parameter	Specification
Sync Pattern	- Programmable length and pattern (8, 16, 24, 32, 40, 48, 56, 64 or 128 bits) - Option to bypass Frame Sync.
Frame Size	16 bytes to 1MBytes
Frame Sync strategy	- Programmable Search, Verify & Lock - SYN- 0 to 15 errors - CTL 0 to 15 frames - LTS 0 to 15 frames - Bit-slip window support – 0, 1, or 2 bits.



De-Randomizer	- CCSDS compliant De-randomizer – programmable for seed word, and polynomial length (7 to 15) - Option to bypass De-Randomization.
OUTPUT SIGNAL AND CONNECTOR	
Parameter	Specification
Supported Output Logic	Simultaneous ECL Differential and LVDS.
ECL Differential	Adaptive Termination V+> -1V, V- < -1.7V
LVDS	100 Ohm differential Termination, Vcm = 1.2V, Amplitude – 400mVpp.
Output Connector Type	2 sets(channels) of 8 SMA(F) for ECL Differential 2 RJ45 connectors for LVDS (4 differential wire pairs available in both case)
Input Connector Type	8 SMA(F) for ECL Differential
Data Format	Selectable from NRZ-L, NRZ-M, NRZ-S.
Data/Clock Signal	- BPSK (Clock and I Data – Max 500Mbps) - QSPK (Split Mode - I Data, Q Data, I Clock, Q Clock: Max 2000Mbps, Merged Mode – I+Q Data and Merged Clock: Max 480MSps) 8PSK (Split Mode in 3 data streams I/Q/Z/Clock – Max 2000Mbps, Split Mode in 2 data streams/Clock– Max 480MSps, Merged Mode I+Q+Z and Merged Clock– Max 480MSps)
Output Clock Duty Cycle	50% nominal and <20ppm - Holdover mode, when received signal is not present or not locked
POST PROCESSING	
Parameter	Specification
Processing Module (Decommuration)	RAW, CADU, VCDU Level
Data Time-Tagging	IRIG-B or External NTP Server, Optional Support of PTP, Optional Support for White-Rabbit protocol for sub-ns accuracy, UTC Target format
CRC Checking	Programmable, 16 bit
CCSDS Decoding	Reed-Solomon, LDPC



Statistics	Pre-FEC BER, Code-word Failure Count, FER
Data-Storage	12TB hard-disk support – High Speed NVMe harddisk to be able to record/playback full throughput data – (4TB * 3)
INBUILT MODULATOR	
Parameter	Specification
Symbol rate	10 - 1000 MSps
Channel	- BPSK –Single Channel - OQPSK/QPSK- Two Channel 8PSK- 3 Channel
Note – Detailed Modulator specification is given in separate table.	
BIT ERROR RATE MEASUREMENT	
Parameter	Specification
BER measuring Range	10^{-3} to 10^{-12} with PRBS sequence
PRBS Mode	PRBS7, PRBS9, PRBS11, PRBS15, PRBS23, PRBS31
	Automatic Delay adjustment between transmitting & received signal, Bit to Bit comparison between transmitted & received signal.
MONITORING AND CONTROL	
Parameter	Specification
Measurements	Instantaneous display of RSSI, Eb/NO, SNR, Received Carrier Offset (Doppler), Bit Clock Offset, BER
Display	Real Time Spectrum, Received Constellation, Digital Equalizer
Control	IF Input Settings, Demod- Settings, Bit-Sync Setting, Decoding Settings, Frame Sync Setting, Output Mode Setting Modulator Setting, etc.
Configuration	Either locally through front panel GUI or remotely through TCP/IP based GUI.
Mission Mode	Option to save configuration in file and directly loading it.
Configuration	Either locally through front panel GUI or remotely through TCP/IP based GUI



OTHER SPECIFICATIONS	
Parameter	Specification
Reference Clock	Support Internal as well as External clock (10 MHz)
Ethernet Ports	Dual Giga-bit Port, 1 10G SFP+ Port for 2Gbps speed over Ethernet
Size	19" Rack Mountable Chassis.
Operating Temperature	25 ± 15° C
Storage Temperature	-10° C to 60° C
Humidity	90%, non-condensing
Power Supply Voltage	230 V ± 10 %, 50 ± 3 Hz

Dual-Channel Built-In Modulator Specification

IF OUTPUT	
Parameter	Specification
No of IF Outputs	2
Input Frequency	720MHz, 2.4GHz
Max Symbol Rate @ 720MHz	320MSps
Max Symbol Rate @ 2.4GHz	1000MSps
IF Tunability @ 720MHz	500MHz to 940MHz
IF Tunability @ 2.4GHz	1.6GHz to 3.15GHz
IF Tunability	±100 KHz step size
Power	-10dBm to -70 dBm in step of 0.1dB
VSWR	≤ 1.5
IF Connector Type	SMA Female
In Band Spurious Level	-40dBc



IF Connector Type	SMA Female (50 ohm)
Note 1: Both IF channels can operate independently and parallelly without any degradation in Output EVM performance. Both channels can select any IF frequency. Note 2: The Max Symbol Rate can be upgraded to up-to 500MSps symbol rate @ 720MHz and 800MSps symbol rate @ 2.4GHz on the same HW with SW only upgradability.	
MODULATION	
Supported schemes	BPSK, QPSK, UQPSK, OQPSK, 8PSK (Selectable)
Spectral Inversion support	I&Q Swap, $\bar{I}\bar{Q}$, $\bar{I}Q$, $I\bar{Q}$
Symbol Rate	10 - 1000 MSPS
Max Bit rate @720MHz	- BPSK (1 X 320) Mbps - QPSK/UQPSK/OQPSK- (2X320) Mbps 8PSK (3 X 320) Mbps
Max Bit rate @2.4GHz	- BPSK (1 X 1000) Mbps - QPSK/UQPSK/OQPSK-(2X1000) Mbps, 8PSK (3 X 1000) Mbps
Pulse Shaping Filter	RRC with Alpha from 0.05 to 0.35 (Programmable), Programmable FIR
Modulation Enable	Selectable (Enable/Disable)
DECODING	
Parameter	Specification
Convolution Encoding	- Constraint length: 7 with polynomial (171, 133)₈ - Selectable rate: 1/2 punctured to 2/3, 3/4, 5/6, and 7/8 - Supports maximum data rate as per hardware capability - Option to enable or disable convolution encoder
Differential Encoder	- Max symbol rate – 1000MSps - Option of Enable/Disable Differential Encoding
RS Encoder	- Compliant to CCSDS recommendation (as per CCSDS 131.0- B-2 Aug 2011). Following RS Encoder are supported. (i) CCSDS RS (223,255) (ii)CCSDS RS (239,255) - Optional Support for CCSDS RS (247,255) - Option of shortening (Virtual fill up-to 16 bytes) and Interleaving depth of 1 to 8 Code-words. - Max Data Rate of 2000Mbps - Option of Enable/Disable RS Encoding



LDPC Encoder	- Compliant to CCSDS 131.1-O-2 - Code-Rate of $\frac{1}{2}$, $\frac{2}{3}$, $\frac{4}{5}$, $\frac{7}{8}$ (i) $\frac{7}{8}$ – Shortened (8160, 7136) code (Code-Rate $\frac{7}{8}$ supports highest data-rate) (ii) $\frac{1}{2}$ - 1K, 4K block length (iii) $\frac{2}{3}$ – 4K block length (iv) $\frac{4}{5}$ – 4K block length - Option of Enable/Disable LDPC Encoding Note - code rate of $\frac{1}{2}$, $\frac{2}{3}$, $\frac{4}{5}$ available as SW upgradation option.
8PSK 4D/6D TCM Encoder	- Output rate 2, 2.25, 2.5 and 2.75 - Option of Enable/Disable TCM Encoding
Data Source for Encoding	External Data Source or Internal Data Source, Each Encoding Scheme can be selectively turned on or turned off.
FRAMING, FRAME SYNC INSERTION & RANDOMIZATION	
Parameter	Specification
ASM/Sync Pattern	- Compliant to CCSDS 131.0-B-2 - Programmable length and pattern (8, 16, 24, 32, 40, 48, 56, 64 or 128 bits) - Option to bypass Frame Sync Insertion. (Can be programmed independently for both the channels)
Frame Size	16 bytes to 1MBytes
Frame Error Control Field Insertion	CCSDS 100.0-G-1, or custom
Randomizer (Scrambling)	- CCSDS 131.0-B-2 compliant Randomizer – programmable for seed word, and polynomial length (7 to 15) - Option to bypass Randomization.
Framing	Frame consists of ASM, User Data, and Parity, and FECF. User File is sent as User Data.
INPUT SIGNAL AND CONNECTOR	
Parameter	Specification
Supported Input Logic	Simultaneous ECL Differential and LVDS.
ECL Differential	Adaptive Impdance $V_{+} > -1V$, $V_{-} < -1.7V$



LVDS	• 100 Ohm differential Impedance, Vcm = 1.2V, • Amplitude – 400mVpp.
Input Connector Type	• 2 sets(channels) of 8 SMA(F) for ECL Differential • 2 RJ45 connectors for LVDS (4 differential wire pairs available in both case)
External Input Data Format	• Selectable from NRZ-L, NRZ-M, NRZ-S.
External Data/Clock Signal	• Both Serial and Parallel Modes Supported • BPSK (Clock and I Data – Max 500MSPS) • QSPK (Split Mode - I Data, Q Data, I Clock, Q Clock: Max 500MSps, Merged Mode – I+Q Data and Merged Clock: Max 480MSps) • 8PSK (Split Mode in 3 data streams I/Q/Z/Clock – Max 500MSps Split Mode in 2 data streams/Clock– Max 480MSps, Merged Mode I+Q+Z and Merged Clock– Max 480MSps)
Data Source to Modulator	• 1. Internal PRBS data source for PRBS 7, 11, 15, 23, 31 • 2. External – Data and Clock Signals in Serial and Parallel Modes on ECL or LVDS signaling. • 3. Binary Files Playback from Internal or External Storage • Data compatible with CCSDS and Binary Raw CADU in User data format.
INBUILT CHANNEL EMULATION	
Parameter	Specification
Symbol rate	• 10 - 1000 MSps
Noise Source Emulation	• AWGN Noise, accurate to 6.6σ • Noise Insertion in full Sampling bandwidth of 500MHz. • Option of On/Off
Doppler Source	• Emulates Satellite Movement • Doppler shift: Up-to 10MHz • Doppler Rate Up-to 50KHz/sec. • Option of On/Off
Path Loss Insertion	• Emulation from 0dB to 50dB.
Fading Tap Emulation	• Support of MIMO Channel Fading Tap Emulation – which make it possible to introduce Cross Polarization Interference effect.



MONITORING AND CONTROL	
Parameter	Specification
Control	• IF Output Settings, Modulator- Settings, Framing Setting, FEC Encoding Settings, Input Mode Setting, Modulation Setting, Channel Emulator Setting, etc.
Configuration	• Either locally through front panel GUI or remotely through TCP/IP based GUI.
Mission Mode	• Option to save configuration in file and directly loading it.
Configuration	• Either locally through front panel GUI or remotely through TCP/IP based GUI.
LICENSING	
Parameter	Specification
Base Unit + Base License	• • Include all • features except below features •
LDPC Decoding	• Optional License
DVB-S2/ DVB-S2x Decoding	• Optional License
SCCC Decoding	• Optional License
2Gbps Speed	• Optional License
Built-In Modulator/Channel Emulator	• Optional License

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Delivering Advanced Solutions for
Next-Gen Networking



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